# Quality Rubrics

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

## One current rubric record with earlier evaluation-rate review preserved.

Explain anchors, apply criteria, reject score compensation and design a revealing rating case.

**Keep one habit:** New answer fields preserve older calculation responses; keep rubric version, evidence and release rule together.

MAIN LESSON · SECTION 1

## Recall: reconnect earlier ideas

> Retrieve before opening notes.

### Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

**My remembered idea, example and correction**

**My remembered idea, example and correction**

▼ **Earlier wording, alternatives and picture transcripts**

**Prototype a Grounded Assistant**

**Requests Responses and JSON**

**User Need and AI Contribution**

# W1 · Explain observable criteria

> Current rubric tasks use separate fields; earlier evaluation-rate answers remain preserved.

## Current source and rubric

Current rubric: source support for every factual claim, answer the asked role question, at most two sentences. All three required; support mandatory. S1 says only A root anchors a plant in soil. Main draft adds an unsupported daily 2 cm growth sentence; removing that clause repairs the support failure. No real model, usability study or broad safety measurement is supplied.

## Current question and response

- Name the three current criteria and their observable anchors. Explain why a count of passed checks cannot override the mandatory support gate.

**My rubric decisions and evidence**

<details>
<summary>▼ Review · Earlier evaluation-rate task and saved response</summary>

Reference for this exercise: open this week's student lesson, "Quality Rubrics", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.
</details>

**My reasoning and evidence**

MAIN LESSON · SECTION 13

# W2 · Apply the three anchors

> Current rubric tasks use separate fields; earlier evaluation-rate answers remain preserved.

## Current source and rubric

Current rubric: source support for every factual claim, answer the asked role question, at most two sentences. All three required; support mandatory. S1 says only A root anchors a plant in soil. Main draft adds an unsupported daily 2 cm growth sentence; removing that clause repairs the support failure. No real model, usability study or broad safety measurement is supplied.

## Current question and response

- Use S1: A root anchors a plant in soil. Question: what role does a root have? Draft: A root anchors a plant in soil. Roots grow exactly 2 cm each day. Apply source support, task fit and at-most-two-sentence format separately, giving evidence for each. Apply the declared all-three release rule, then predict removing only the unsupported sentence.

**My rubric decisions and evidence**

▼ **Review · Earlier evaluation-rate task and saved response**

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

---

MAIN LESSON · SECTION 14

# W3 · Reject a score-over-gate argument

> Current rubric tasks use separate fields; earlier evaluation-rate answers remain preserved.

## Current source and rubric

Current rubric: source support for every factual claim, answer the asked role question, at most two sentences. All three required; support mandatory. S1 says only A root anchors a plant in soil. Main draft adds an unsupported daily 2 cm growth sentence; removing that clause repairs the support failure. No real model, usability study or broad safety measurement is supplied.

## Current question and response

- A draft passes task fit and length but contains an unsupported measurement. Can a two-of-three score justify release under the stated all-three rule? Explain what must change.

**My rubric decisions and evidence**

▼ **Review · Earlier evaluation-rate task and saved response**

A report omits this qualification: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

MAIN LESSON · SECTION 16

# W5 · Design a revealing rubric check

> Current rubric tasks use separate fields; earlier evaluation-rate answers remain preserved.

## Current source and rubric

Current rubric: source support for every factual claim, answer the asked role question, at most two sentences. All three required; support mandatory. S1 says only A root anchors a plant in soil. Main draft adds an unsupported daily 2 cm growth sentence; removing that clause repairs the support failure. No real model, usability study or broad safety measurement is supplied.

## Current question and response

- Propose a fully supplied draft that passes one criterion and fails another under S1. Predict its flags and release decision. Then improve one ambiguous anchor and explain why all drafts must be rescored under the revised rubric.

**My rubric decisions and evidence**

## ▼ Review · Earlier evaluation-rate task and saved response

Reference for this exercise: open this week's student lesson, "Quality Rubrics", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

# W4 · Judge the seed-coat draft independently

> Complete before hints; record support used.

## Current independent task and response

- Approved C2 says: A seed coat protects the seed. Question: what role does the seed coat have? Draft: A seed coat protects the seed. Every seed opens exactly 48 hours later. Use the same three criteria: every factual claim supported by C2, answer the asked role question, at most two sentences. All three are required for release.

  Give a separate judgement with evidence for each, the release decision and one justified repair.

> **My rubric decisions and evidence**

▼ **Review · Earlier rate assessment and saved response**

### Independent case: Apply a new case independently

Twelve cases: eight answerable, four unanswerable. Correct answers six; appropriate abstentions two. Compute separate and combined pass rates.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

---

MAIN LESSON · SECTION 9

## Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

### Your evidence record

Use a five-**dimension** rubric and block release when correctness or safety is below threshold.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

---

MAIN LESSON · SECTION 17

## Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

### Your evidence record

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

---

MAIN LESSON · SECTION 10

## Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

### Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

### Use this when helpful

Earlier evaluation-rate completion practice: preserve the task and saved response as review.
Use current W2 for this week's supported quality-rubric practice.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

## Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

### Use this when helpful

Use the worked case for System evaluation sets. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

## Reference · Original learning target

> Use this target to inspect an explanation.

### Use this when helpful

An answer earns 2 for format, 2 for tone and 0 for source support. It gets 4/6. Is it acceptable for a source-only assistant? Revise the rubric's decision rule.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

## Extra practice · A complete case

> Try the question before checking the worked explanation.

### Use this when helpful

Rubric gives 0–2 each for support, usefulness and safety. Response is useful and safe but unsupported.

Why can a total mislead?

Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

## Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

### Use this when helpful

Two answers have equally polished language. One has three supported claims; the other has two supported claims and one invented claim. How should a quality rubric distinguish them?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

**Week 25 Test and reflect**

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Predict before tracing or testing.

### Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ Reference · Original target response space

REFERENCE · SECTION 11

# Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

## Use this when helpful

An answer earns 2 for format, 2 for tone and 0 for source support. It gets 4/6. Is it acceptable for a source-only assistant? Revise the rubric's decision rule.

### My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier evaluation-rate task

REFERENCE · SECTION 15

# Additional practice · Earlier evaluation-rate task

> This retained calculation differs from the current rubric assessment.

## Earlier task and response

- Twelve cases: eight answerable, four unanswerable. Correct answers six; appropriate abstentions two. Compute separate and combined pass rates.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

# Extra practice · Use the worked picture

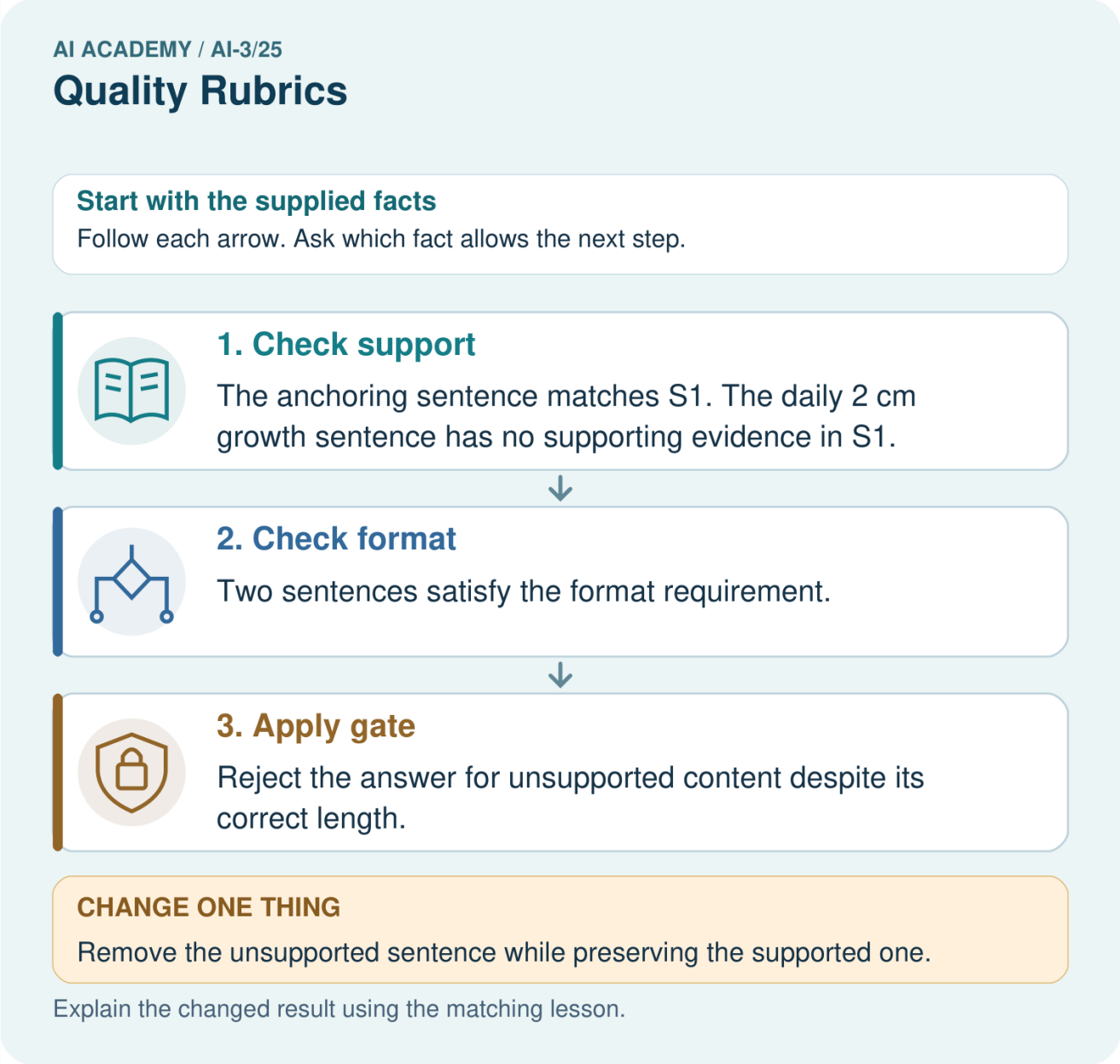
> Keep the explanation and the support you used.

## Use this when helpful

Use a quality rubric whose safety gate cannot be traded for attractive formatting.

Approved S1 says: “A root **anchor**s a plant in soil.” Rubric: every claim supported by S1, at most two sentences. Draft: “A root **anchor**s a plant in soil. Roots grow exactly 2 cm each day.”

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.




Follow the labelled picture with the matching explanation. Open the vector diagram

Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

# Reference · Project portfolio record

> Extend your main project response when helpful.

## Use this when helpful

Project: Verified Study Buddy

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- Use a quality rubric whose safety gate cannot be traded for attractive formatting.
- Which labelled step matches this detail? Reject the answer for unsupported content despite its correct length.
- 1. Check format

- 2. Apply gate
- 3. Check support
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Apply gate Reject the answer for unsupported content despite its correct length. Compare each draft sentence directly with S1.

  The **anchor**ing claim matches the supplied source; the daily growth claim adds a measurement the source never gives. Two sentences pass the format rule, but unsupported content fails the mandatory **evidence** gate.

  Removing that sentence repairs the support defect rather than merely polishing the wording.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

### Practice 2: explain the deciding evidence

## Changed-case question

- Changed-case practice: Remove the unsupported sentence while preserving the supported one. Which conclusion is supported by the supplied facts?
- 1. The revised one-sentence answer passes these two declared checks.

- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The revised one-sentence answer passes these two declared checks. The remaining claim is supported and one sentence is within the maximum.

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check the changed case**

**Repair a missing building block**

## Repair and save your work

Review Prototype a Grounded Assistant (ai-3/24). Goal: Explain the **mechanism** and terminology of capstone grounded assistant. Short practice: Can PL2 alone support “check the soil daily”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Keep your completed work**

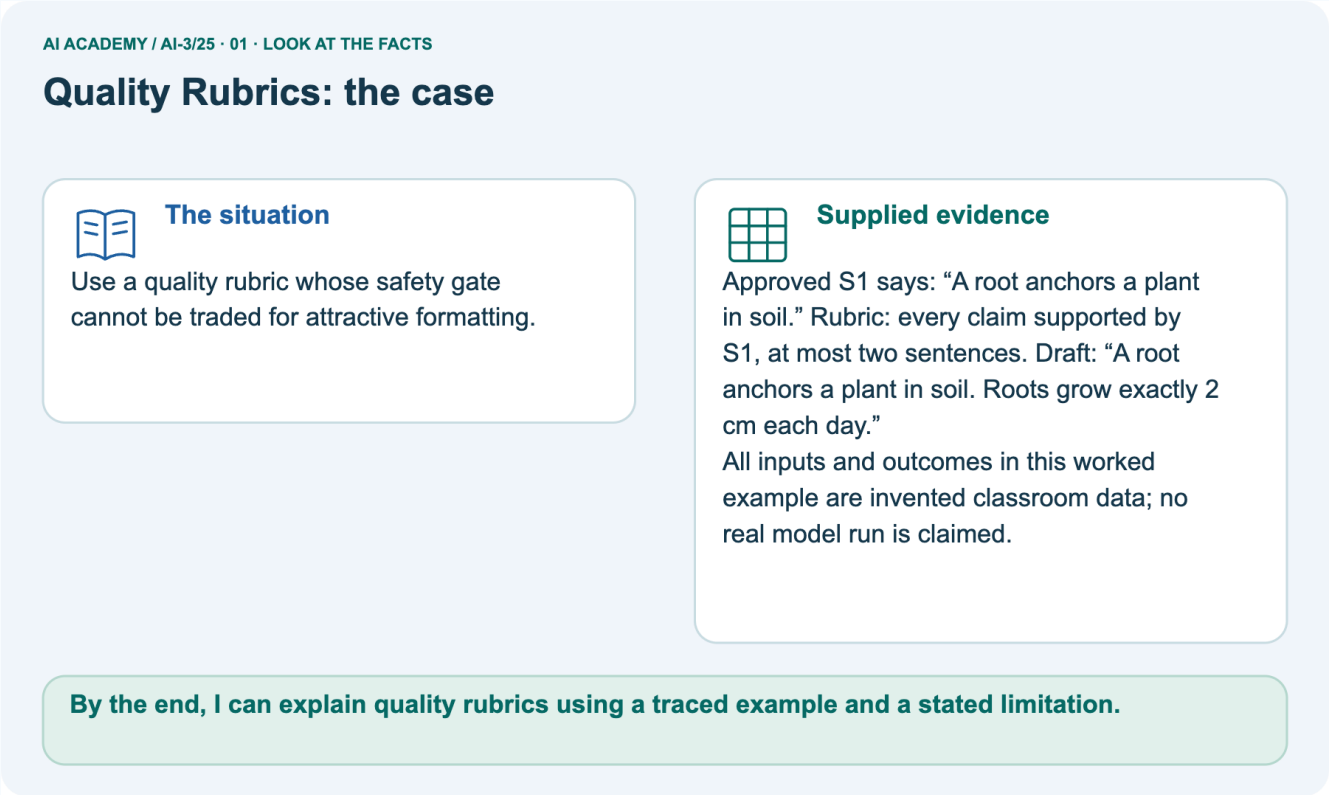
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

## 01. Quality Rubrics: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

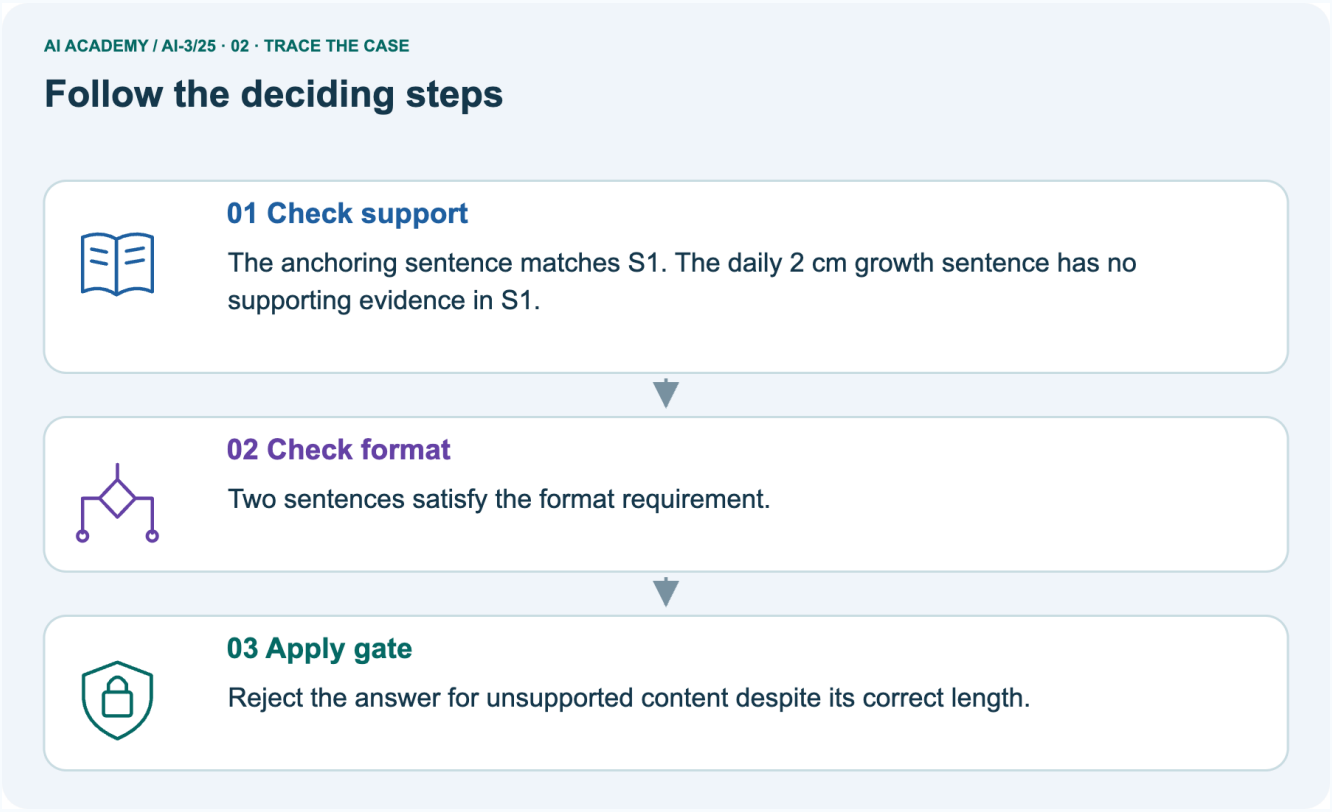
Use a quality rubric whose safety gate cannot be traded for attractive formatting. Supplied facts: Approved S1 says: “A root **anchor**s a plant in soil.” Rubric: every claim supported by S1, at most two sentences.

Draft: “A root **anchor**s a plant in soil. Roots grow exactly 2 cm each day.” All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

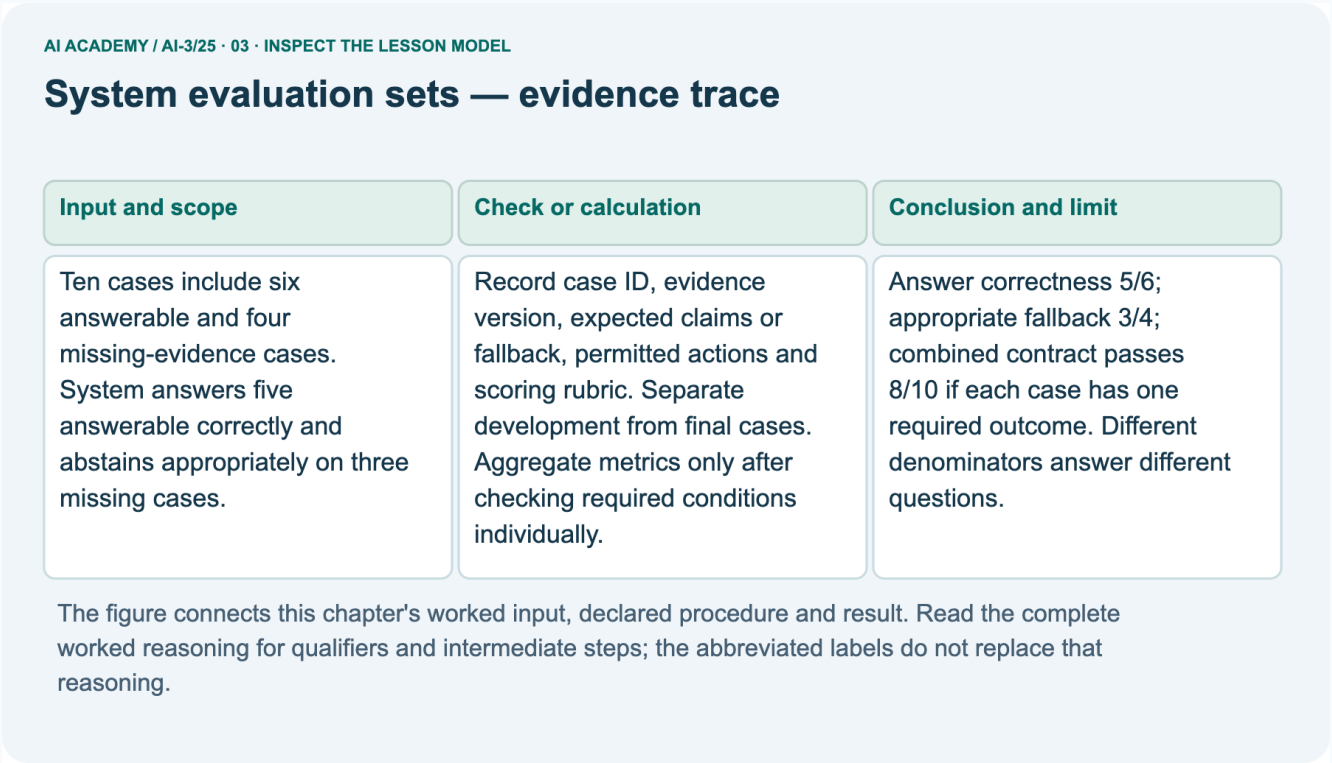
Check support: The **anchor**ing sentence matches S1.

The daily 2 cm growth sentence has no supporting **evidence** in S1. → Check format: Two sentences satisfy the format requirement. → Apply gate: Reject the answer for unsupported content despite its correct length.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. System evaluation sets — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

System evaluation sets — **evidence** trace. Ten cases include six answerable and four missing-**evidence** cases.

System answers five answerable correctly and abstains appropriately on three missing cases. | Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

Separate development from final cases.

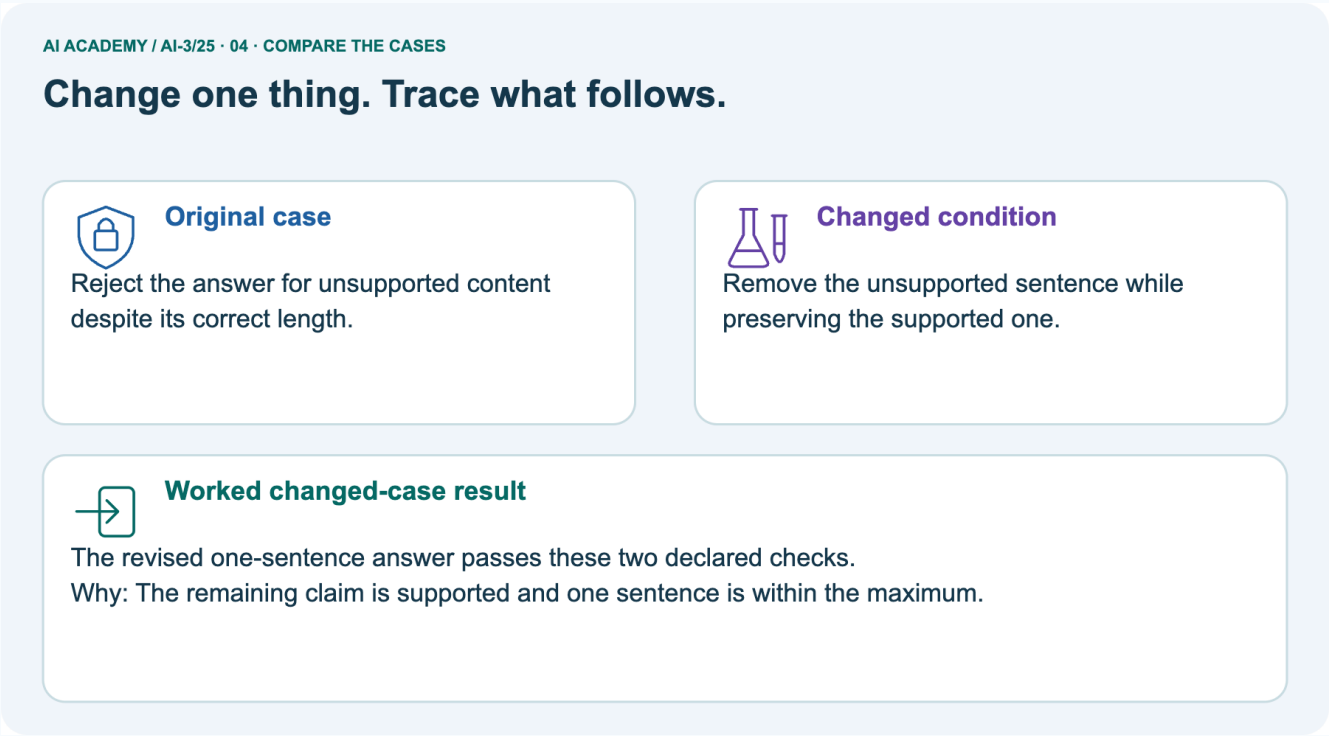
Aggregate metrics only after checking required conditions individually. | Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome.

Different denominators answer different questions.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

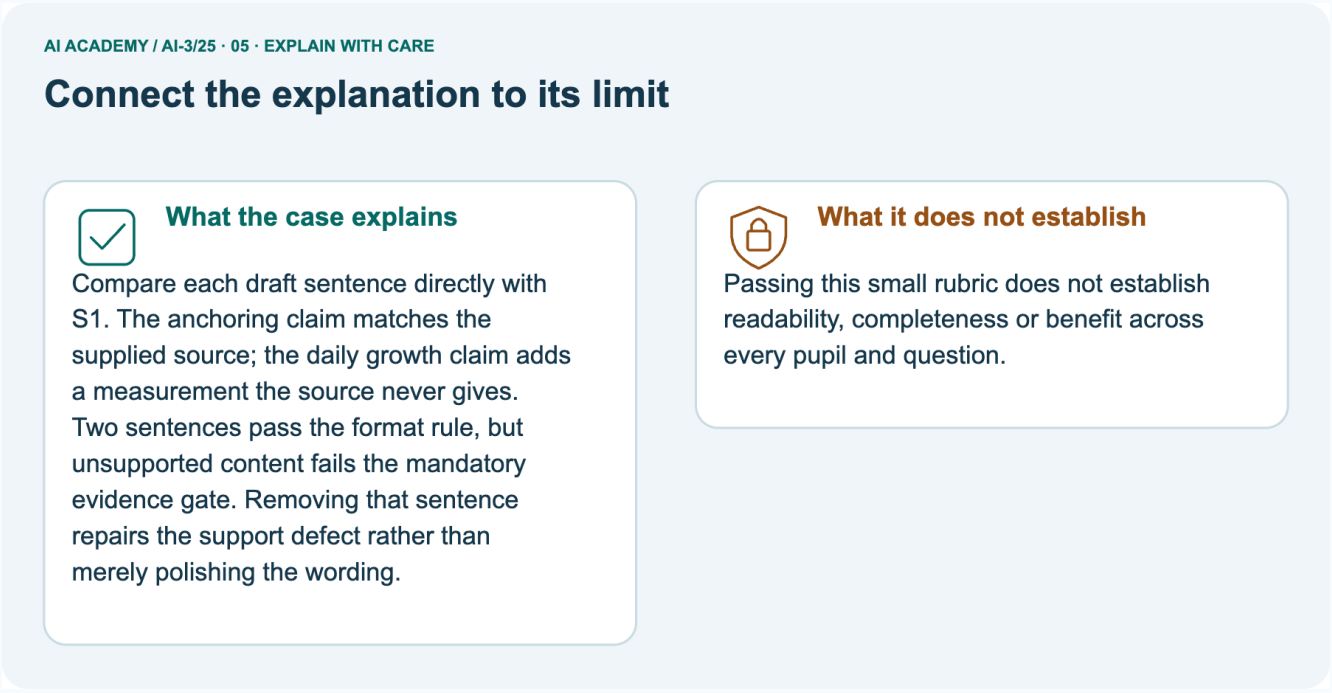
Original case: Reject the answer for unsupported content despite its correct length. Changed condition: Remove the unsupported sentence while preserving the supported one.

Expected result: The revised one-sentence answer passes these two declared checks.
Reason: The remaining claim is supported and one sentence is within the maximum.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Compare each draft sentence directly with S1. The **anchor**ing claim matches the supplied source; the daily growth claim adds a measurement the source never gives.

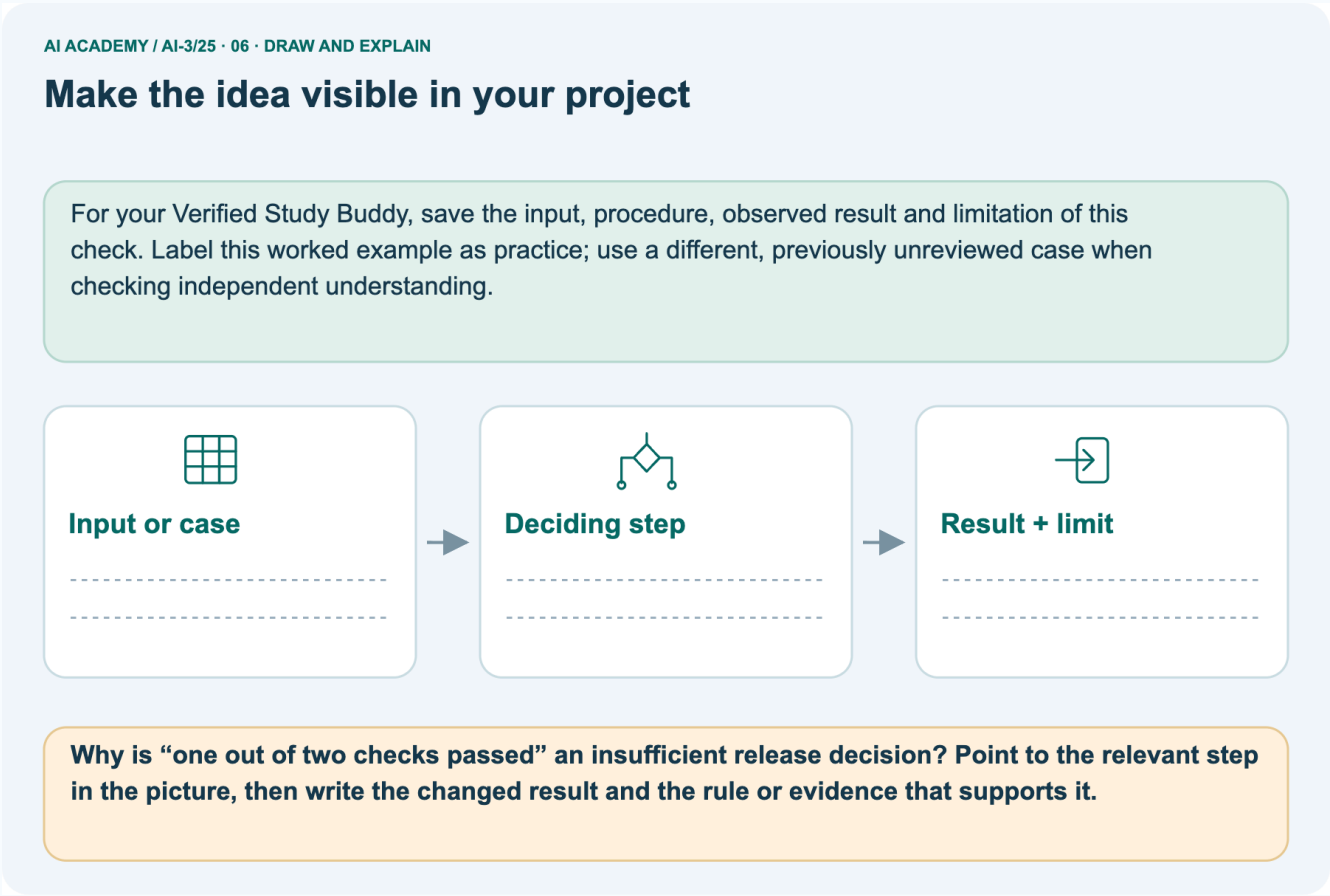
Two sentences pass the format rule, but unsupported content fails the mandatory **evidence** gate. Removing that sentence repairs the support defect rather than merely polishing the wording.

Limit: Passing this small rubric does not establish readability, completeness or benefit across every pupil and question.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

**01. Quality Rubrics: the case**

**02. Follow the deciding steps**

**03. System evaluation sets — evidence trace**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.